

```
from google.colab import files

uploaded = files.upload()
```

[Choose Files](#) | Hotel_Dataset.xlsx
Hotel_Dataset.xlsx(application/vnd.openxmlformats-officedocument.spreadsheetml.sheet) - 5733 bytes, last modified: 9/16/2026 - 100% done
Saving Hotel_Dataset.xlsx to Hotel_Dataset (2).xlsx

```
import pandas as pd

df = pd.read_excel('/content/Hotel_Dataset (2).xlsx')
df.to_csv('/content/Hotel_Dataset.csv', index=False)
```

```
files.download('/content/Hotel_Dataset.csv')
```

```
import pandas as pd

df = pd.read_csv('/content/Hotel_Dataset.csv')
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1	
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25	
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35	
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30	
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25	
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+	
5	6	35+	3	lbys	Non-Veg	1909	2	122220	35+	
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+	
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25	
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30	
9	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30	
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35	

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.duplicated()
```

```
0
0 False
1 False
2 False
3 False
4 False
5 False
6 False
7 False
8 False
9 True
10 False
```

dtype: bool

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
#   Column          Non-Null Count  Dtype
---  -
0   CustomerID      11 non-null    int64
```

```
1 Age_Group      11 non-null    object
2 Rating(1-5)    11 non-null    int64
3 Hotel          11 non-null    object
4 FoodPreference 11 non-null    object
5 Bill           11 non-null    int64
6 NoOfPax        11 non-null    int64
7 EstimatedSalary 11 non-null    int64
8 Age_Group.1    11 non-null    object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

```
df.drop_duplicates(inplace=True)
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	lbys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
index = np.array(list(range(0, len(df))))
df.set_index(index, inplace=True)
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	lbys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.drop(['Age_Group.1'], axis=1, inplace=True)
df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	Ibys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.loc[df.CustomerID < 0, 'CustomerID'] = np.nan
df.loc[df.Bill < 0, 'Bill'] = np.nan
df.loc[df.EstimatedSalary < 0, 'EstimatedSalary'] = np.nan

df.loc[(df.NoOfPax < 1) | (df.NoOfPax > 20), 'NoOfPax'] = np.nan

df.loc[(df['Rating(1-5)'] < 1) | (df['Rating(1-5)'] > 5), 'Rating(1-5)'] = np.nan

df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	NaN	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	NaN	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4.0	87777.0

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.Age_Group.unique()

array(['20-25', '30-35', '25-30', '35+'], dtype=object)

df.Hotel.unique()

array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)

df.FoodPreference.unique()

array(['veg', 'Non-Veg', 'Veg', 'Vegetarian', 'non-Veg'], dtype=object)

df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)

df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
df.FoodPreference.replace(['non-Veg'], 'Non-Veg', inplace=True)

df
```

/tmp/ipykernel_1930/3263569114.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df.Hotel.replace(['Ibys'], 'Ibis', inplace=True)
```

/tmp/ipykernel_1930/3263569114.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

```
df.FoodPreference.replace(['Vegetarian', 'veg'], 'Veg', inplace=True)
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0	
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0	
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2.0	30000.0	
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2.0	120000.0	
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0	
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0	
6	7.0	35+	4.0	RedFox	Veg	1000.0	NaN	21122.0	
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	NaN	345673.0	
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	NaN	
9	10.0	30-35	5.0	RedFox	Non-Veg	NaN	4.0	87777.0	

Next steps:

[Generate code with df](#)
[New interactive sheet](#)

```
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)
df.Bill.fillna(round(df.Bill.mean()), inplace=True)
```

```
df.NoOfPax.fillna(round(df.NoOfPax.median()), inplace=True)
df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)
```

```
df
```

/tmp/ipykernel_1930/2651257644.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c...
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()), inplace=True)

-----/tmp/ipykernel_1930/2651257644.py:2:- FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c...
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

Next steps: [Generate code with df](#) [New interactive sheet](#)

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

df.info()
df

/tmp/ipykernel_1930/2651257644.py:4: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through c...
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are

Data columns (when doing df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col]

#	Column	Non-Null Count	Dtype
0	CustomerID	10 non-null	float64
1	Age_Group	10 non-null	object
2	Rating(1-5)	10 non-null	float64
3	Hotel	10 non-null	object
4	FoodPreference	10 non-null	object
5	Bill	10 non-null	float64
6	NoOfPax	10 non-null	float64
7	EstimatedSalary	10 non-null	float64

dtypes: float64(5), object(3)
memory usage: 728 bytes

0	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	20-35	4.0	LemonTree	Non-Veg	2022.0	3.0	50000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-35	4.0	LemonTree	Veg	1089.0	2.0	14000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	RedFox	Non-Veg	1000.0	2.0	12222.0
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-35	4.0	LemonTree	Non-Veg	2400.0	3.0	34000.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	1801.0	4.0	87777.0